

# **CHICAGO CRIME PROBABILITY PREDICTION PROJECT**

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# CHICAGO CRIME PROBABILITY PREDICTION PROJECT

- • Predict crime occurrence probability in Chicago.
- • Leverages machine learning for crime risk estimation.
- • Includes extensive exploratory data analysis (EDA).
- • Interactive web app developed using Streamlit.
- • Supports real-time decision-making for public safety.

# PROBLEM STATEMENT

- Aim: Estimate likelihood of a crime happening.
- Useful for policing and community safety strategies.
- Identify risk factors like time, location, and category.
- Assist in resource planning and safety initiatives.
- Provide a predictive view of crime dynamics.

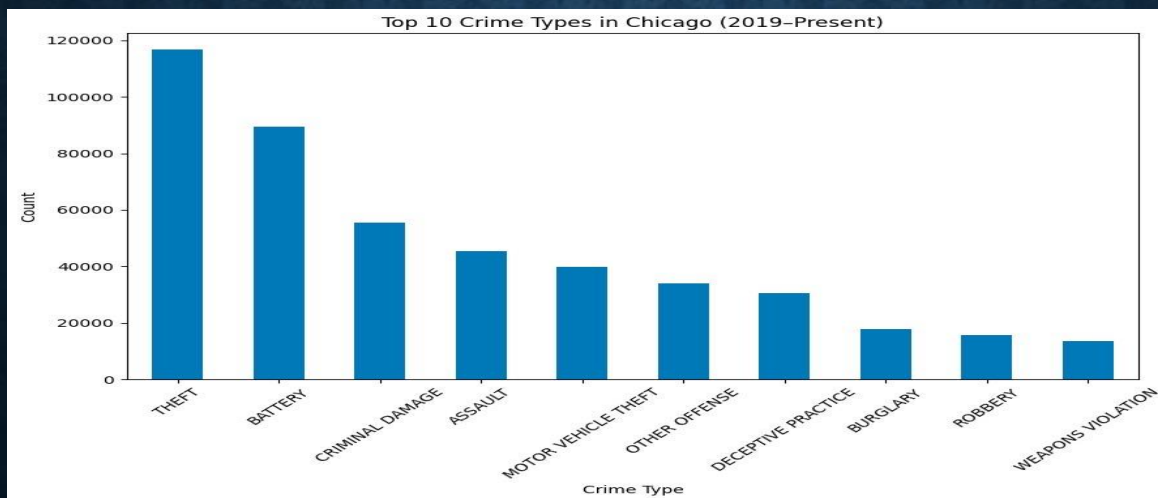


# **DATASET OVERVIEW**

- Data sourced from Chicago Data Portal (2019–2025).
- Includes crime type, location, time, and coordinates.
- Categorized as domestic vs non-domestic crimes.
- Dataset used for both EDA and machine learning.
- Large and diverse dataset enhances model accuracy.

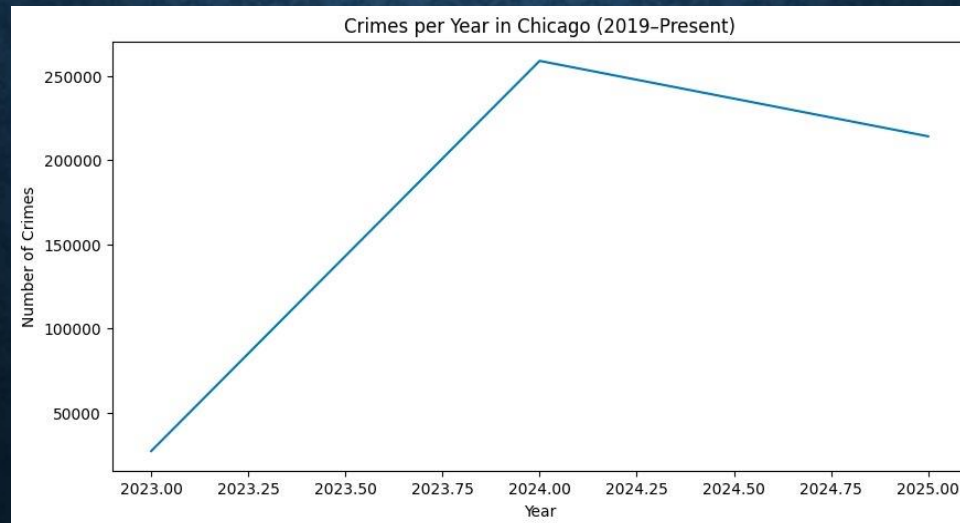
# CRIME TYPE ANALYSIS

- Most common crimes: Theft, Battery, Criminal Damage.
- Theft contributes the highest proportion of crime.
- Crime categories indicate behavioral patterns.
- Helps identify major crime clusters in the city.
- Important for selecting meaningful ML features.



# CRIME TRENDS OVER YEARS

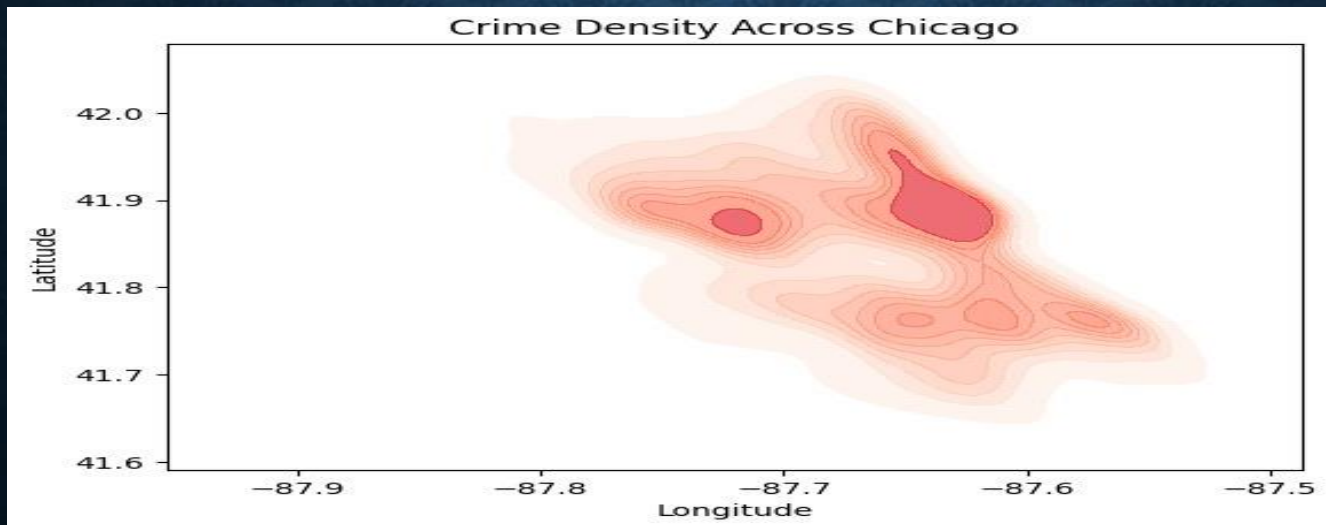
- • Crime trends vary significantly by year.
- • 2024 shows the highest crime frequency.
- • Temporal patterns influence prediction accuracy.
- • Year and month used as essential input features.
- • Trends help understand seasonality in crime rates.





# CRIME DENSITY MAP

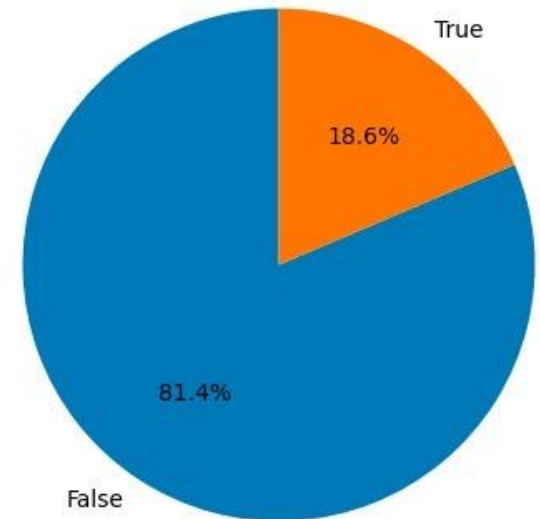
- • Heatmap reveals high-crime hotspots across Chicago.
- • North-east Chicago shows strong clustering.
- • Spatial patterns inform safety policies.
- • Critical for geospatial crime understanding.
- • Supports hotspot-based policing strategies.



# EDA: DOMESTIC VS NON-DOMESTIC CRIMES

- • Non-domestic crimes dominate with ~81%.
- • Domestic crimes form a small proportion (~19%).
- • Category imbalance impacts ML behavior.
- • Highlights differences in crime intent/context.
- • Useful for designing categorical features.

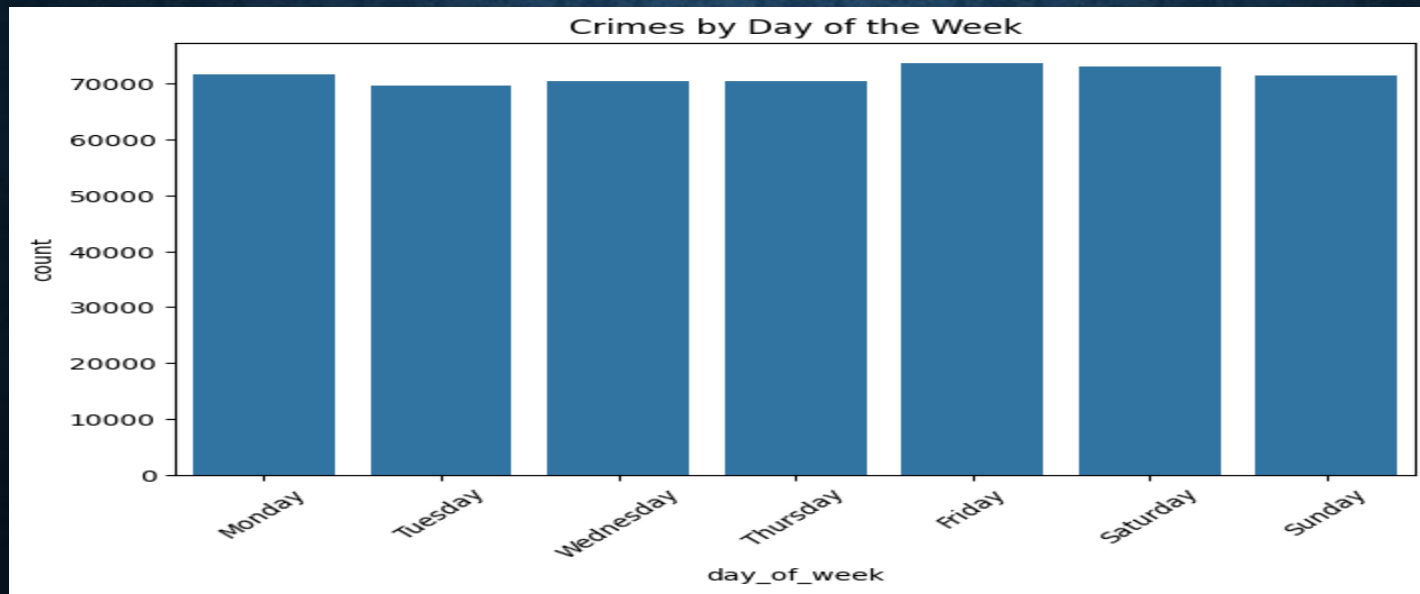
Domestic vs Non-Domestic Crimes





# WEEKLY CRIME PATTERNS

- • Fridays show highest weekly crime activity.
- • Tuesdays and Thursdays have the lowest crime counts.
- • Weekly cycles influence prediction modelling.
- • Important temporal feature for analysis.
- • Supports detection of routine crime behaviors.



# FEATURE ENGINEERING

- • Extracted hour, weekday, month, year from timestamps.
- • Applied One-Hot Encoding to categorical fields.
- • Normalized numerical inputs for consistency.
- • Removed irrelevant/missing attributes.
- • Prepared structured dataset for ML modeling.

# MODEL TRAINING PROCESS

- • Model chosen: Random Forest Classifier.
- • Handles mixed data types effectively.
- • Provides robust probability-based predictions.
- • Performs well with large datasets.
- • Reliable and interpretable ML approach.



# SYNTHETIC NEGATIVE SAMPLING

- • Original dataset contains only crime cases (1).
- • Generated synthetic 'no crime' records (0).
- • Balanced dataset improves classifier fairness.
- • Essential for accurate probability prediction.
- • Prevents bias toward crime-only predictions.

# MODEL EVALUATION

- • Achieved strong classification accuracy.
- • Evaluated using accuracy and probability metrics.
- • Good separation between crime and no-crime cases.
- • Reliable for real-time predictions.
- • Supports practical decision-making in deployment.

# STREAMLIT APP OVERVIEW

- • User-friendly Streamlit interface created.
- • Inputs include time, location, and crime type.
- • Outputs probability score of crime occurrence.
- • Dark-themed UI for clean visualization.
- • Real-time response with ML model integration.



# PREDICTION UI

- • Screenshot showcasing prediction interface.
- • Demonstrates user selections for inputs.
- • Displays crime probability output clearly.
- • Risk levels: Low, Medium, High shown visually.
- • Highlights user interaction workflow.



2 Tabs

- Bookmarks
- Reading List
- Shared with You
- iCloud Tabs



localhost



Chicago Crime Arrest Prediction



Chicago Crime Probability Predictor

Deploy



# Chicago Crime Probability Prediction App

Enter the conditions below to estimate the probability of a crime occurring.

Crime Category

NARCOTICS

Location Description

STREET

Hour of Day

7

Day of Week

Wednesday

Month

1

Year

2021

Predict Crime Probability



## Prediction Result

Estimated Crime Probability: 12.04%



Low Likelihood of Crime

# CONCLUSION & FUTURE SCOPE

- • Predictive model offers actionable crime insights.
- • Future work: add real-time streaming crime data.
- • Could include weather/socio-economic indicators.
- • Potential deployment on cloud for public access.
- • Extend system with deep learning enhancements.